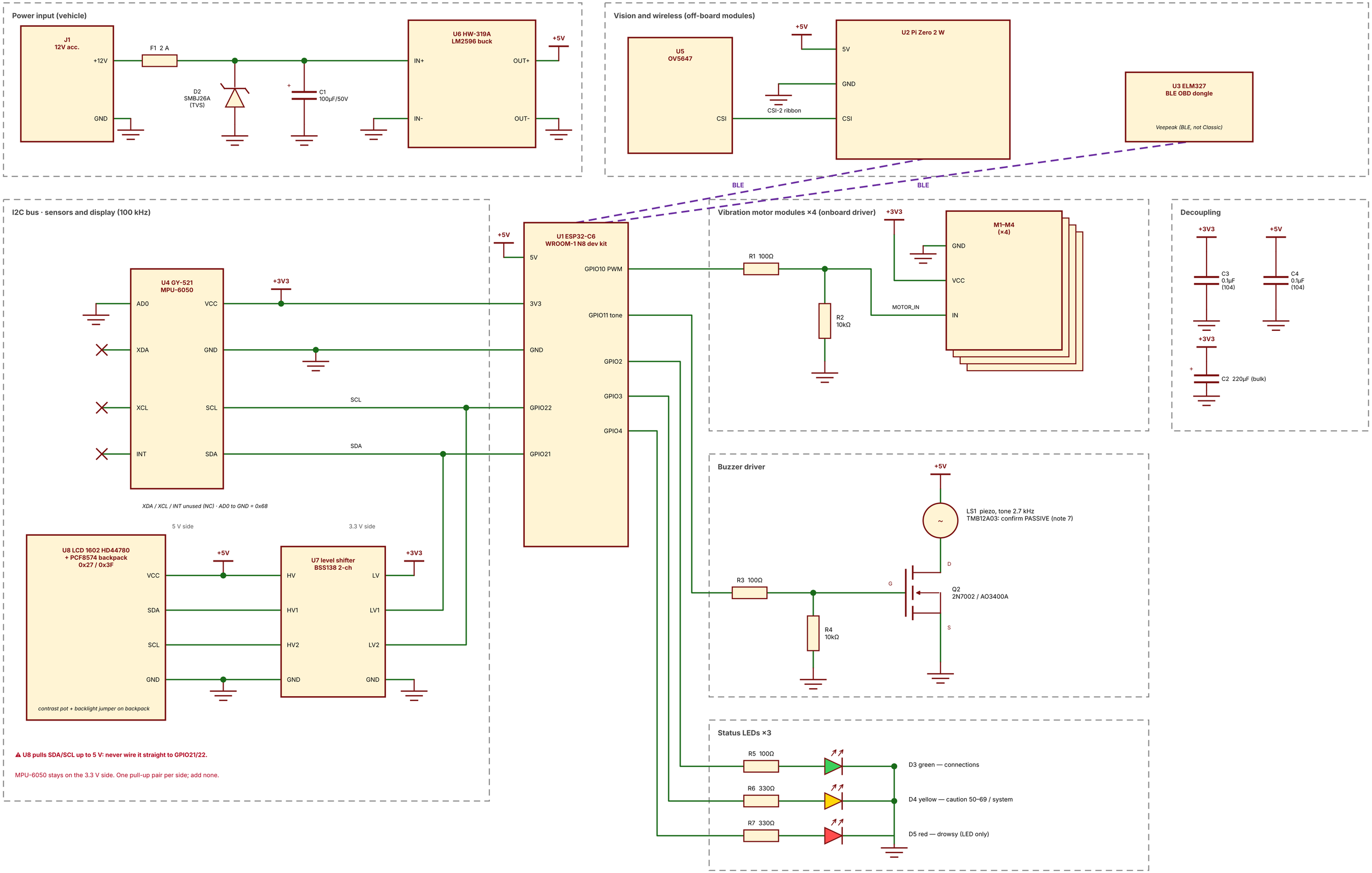
DRIFTS — Schematic (NewBrain architecture) · rev 3

Team 1 · ESP32-C6 dev kit + Raspberry Pi Zero 2 W · rev 3 adds status LEDs ×3, 16×2 I2C LCD and I2C level shifter · generated drawing (not a KiCad file)



NOTES

1. U6 = HW-319A-V7.0.1 LM2596 module. Trim the output to 5.0 V with the trimmer BEFORE connecting the Pi or C6. Bench/demo: skip the 12 V front end and use a USB power bank.
2. Motor modules M1–M4 (GND, VCC, IN) have an onboard driver. IN ← GPIO10 (20 kHz PWM, 8-bit) via R1 100Ω; R2 10k holds IN low during reset.
3. Module VCC drawn from +3V3. Verify the module rating and that the C6 LDO can supply 4 × 60–100 mA plus the IMU; +5V is the alternative.
4. U4 GY-521: AD0→GND = 0x68. Wire.begin(21, 22). Breakout has pull-ups.
5. I2C runs at 100 kHz (PCF8574 limit), shared by U4 and U8. U7 BSS138 shifter: LV = +3V3, HV = +5V. U8 backpack is a 5 V part and pulls SDA/SCL to 5 V: never connect it directly to the C6. U8 address 0x27 (PCF8574) or 0x3F (PCF8574A), auto-detected.
6. LEDs: D3 green GPIO2 (100Ω true green, 330Ω yellow-green), D4 yellow GPIO3, D5 red GPIO4. ~12 mA total. GPIO4 is a strapping pin: an LED to GND only, nothing else. Mount LEDs and LCD in the passenger's view.
7. LS1 must be PASSIVE for tone (GPIO11 → Q2). The fitted TMB12A03 may be ACTIVE: check with TEST:0:3000:DC. No flyback diode for a piezo.
8. C3/C4: 0.1µF ceramic at +3V3 and +5V near U1; one per module VCC.
9. Wireless (dashed purple): Pi↔C6 and OBD dongle↔C6 over BLE. U3 must be BLE, not Classic. Camera U5 connects to the Pi only.
10. Do not use: 5, 8, 9, 15 (strapping); 12, 13 (USB); 16, 17 (UART0 to CH343); 14, 24–30 (flash). GPIO4 = red LED only.
11. Same-named flags (+5V, +3V3, GND) are one net, as in KiCad.
12. Firmware v12: LCD shows INTERVENTION 1/2 (score 50–69 / <50), same bands as the LEDs. See Docs/lcd-display.md, Hardware/pinout.md.